

Ankit Kumar

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PROFESSIONAL SUMMARY

M.Sc. student in Mathematics and Scientific Computing at NIT Warangal with hands-on experience in Data Science, Machine Learning, and AI-driven analytics. Experienced in developing end-to-end ML workflows involving data preprocessing, model optimization, cross-validation, and performance evaluation. Building LLM-powered analytics applications using prompt engineering and natural-language interfaces to transform complex data into actionable, data-driven insights.

EDUCATION

- National Institute Of Technology, Warangal** 2024-2026
Master of Science, Mathematics and Scientific Computing
- Industrial Training Institute, Giridih Pachamba** 2022-2024
National Council for Vocational Training
- St. Columba's College, Hazaribagh** 2019-2022
Bachelor of Science, in Mathematics

PERSONAL PROJECTS

• AI Data Analyst Chatbot

Project Description: LLM-powered conversational analytics platform for querying datasets and generating automated, context-aware insights through natural-language interaction.

- Developed LLM-powered AI data analytics chatbot that enables users to query structured datasets using natural-language questions and receive context-aware analytical responses without manually writing SQL or Python code.
- Engineered an end-to-end analytical workflow integrating natural-language query processing, dataset inspection, data manipulation, statistical analysis, and LLM-based response generation to automate exploratory data analysis and insight discovery.
- Developed intent-driven Plotly visualizations with conversational memory, multi-turn agentic workflows, advanced data validation, conditional analysis, automated insights, export/reporting, and production-ready error handling for end-to-end natural-language data analytics.
- Implemented intelligent analytical capabilities including descriptive statistics, dataset profiling, missing-value analysis, categorical and numerical analysis, trend identification, comparative analysis, and business-oriented insight generation.

• Big Mart Sales Prediction

Project Description: Architected an end-to-end machine learning pipeline to predict product-level retail sales using product, pricing, and store-level attributes for data-driven inventory and marketing decisions.

- Forecasting Pipeline: Built an end-to-end sales prediction pipeline, including robust data cleaning, imputation, and EDA using Pandas and Matplotlib/Seaborn.
- Feature Engineering: Transformed categorical data using One-Hot Encoding and engineered impactful features such as Outlet_Age to improve system performance
- Benchmarked Linear Regression, Ridge, Lasso, Random Forest, and XGBoost using RMSE and R^2 ; Lasso achieved the strongest reported holdout performance with **RMSE 999.13** and **R^2 0.867..**
- Actionable Insights: Leveraged the model to identify Item_MRP and Outlet_Identifier as key sales drivers, providing a quantitative basis for strategic pricing and inventory decisions.

• Walmart Sales Analysis

Analyzed retail sales dataset to uncover revenue trends, seasonal patterns, and customer behavior.

- SQL Database Analysis: Created a data analysis pipeline in MySQL, performing data cleaning and preparation on a large retail dataset. Executed advanced SQL queries using Joins and aggregate functions to analyze sales performance across different stores and departments.
- Trend Identification: Utilized Python with Pandas to conduct an in-depth analysis of sales data, identifying key trends and the impact of holiday promotions and seasonal weather patterns on weekly revenue.
- Data Visualization & Insights: Enhanced a suite of visualizations using Matplotlib to clearly present complex findings, providing actionable insights to support data-driven decisions in inventory management and marketing.

TECHNICAL SKILLS

Programming: Python, SQL, R, C++, Excel, Power BI, MATLAB

Data Analysis: Pandas, NumPy, SciPy, Data Wrangling, Statistical Analysis,

Machine Learning: Scikit-learn, Regression, Classification, Clustering, Dimensionality Reduction, Feature Engineering, Predictive Modeling, Model Training, Model Evaluation, Cross-Validation, Model Optimization, XGBoost

Data Visualization: Matplotlib, Seaborn, Plotly, Interactive Visualization, Dashboard Creation

AI / GenAI: Large Language Models (LLMs), Generative AI, Prompt Engineering, Natural-Language Interfaces, Groq API

Web Data & APIs: Web Scraping, BeautifulSoup, REST APIs, Data Collection

Databases: MySQL, Microsoft SQL Server, Relational Databases, SQL Joins, Views, Stored Procedures, Transactions, DDL, DML

Development & Tools: Jupyter Notebook, JupyterLab, Git, GitHub, RStudio, Pickle

POSITIONS OF RESPONSIBILITY

•**Member** Mathematics Society NITW (2024-2026)

•**Class Representative** St. Columba's College, Hazaribagh (2019-2021)

ACHIEVEMENTS

• IIT-JAM(MA) Qualifier(**AIR Rank-535**) (2024)

• Bal Patrakar (Child Journalist) **UNICEF** , Bal Mitra Akhbar (2014- 2016)
Contributed articles to magazine.

CERTIFICATES

• IBM Data Science Professional Certificate

• Microsoft SQL Server Professional Certificate

• Machine Learning Introduction for Everyone, IBM

• SQL to Work: The TechMart Challenge, Coursera